Code No: 114CX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, September/October - 2023
JAVA PROGRAMMING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	What is Byte code?	[2]
b)	What is the purpose of 'this' keyword in java?	[3]
c)	What is an Inner class?	[2]
d)	Define Abstract class and Abstract method.	[3]
e)	What are the two ways of implementing thread in Java?	[2]
f)	Compare and contrast throw and throws.	[3]
g)	What are the advantages of Collections Framework?	[2]
h)	What is JDBC? List the JDBC components?	[3]
i)	Define Adapter classes.	[2]
j)	What is an Event Listener in java?	[3]

PART - B

(50 Marks)

2.a) Explain any five features of object oriented programming.
b) Write a Program to illustrate static variables and static methods. [5+5]

OR

3.a) What is garbage collection in Java? How it is achieved?
b) Write a Java program to implement constructor overloading. [5+5]

4.a) What is Method overriding? Write a program to demonstrate method overriding.
b) What is a Package? Write down the steps for creating user defined package. [7+3]

OR

5. Explain how interface is used to achieve multiple inheritance in Java? [10]

6. What is an exception? Write a program to accept a password from user and throw "Authentication Denied" exception if the password is incorrect. [10]

OR

7.a) Explain with a neat diagram the life cycle of a thread.
b) Differentiate between an error and an exception. Give examples for the same. [5+5]

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8. Explain JDBC type 1 to type 4 drivers. [10]

OR

9.a) What is a stream? Explain Byte Stream and Character Stream.
b) What is the use of ArrayList class? Describe any two methods of Array List class. [5+5]

10.a) Write a java Program to handle various keyboard events.
b) Explain the lifecycle of an applet. [5+5]

OR

11.a) Compare and contrast Swing and Abstract Window Toolkit (AWT) in Java.
b) Briefly explain Delegation Event Model. [5+5]

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